

Analyzing Student Performance with Machine Learning

A Data Science Approach

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1 Introduction

This academic project explores how machine learning techniques can be applied to predict and understand student performance. Using student profile data including behavioral, academic, and contextual features, I aimed to identify patterns and assess predictive variables related to exam outcomes.

2 Dataset Overview

The primary dataset used includes 6,607 student records with 20 features. These features span:

- Academic Behavior: `Hours_Studied`, `Attendance`, `Previous_Scores`
- Environment and Context: `Motivation_Level`, `Parental_Involvement`, `Family_Income`
- Target Variable: `Exam_Score`

The data was sourced from multiple public student datasets on Kaggle. After evaluating the schemas, similar datasets were combined and standardized for consistency. This included aligning column names, converting categorical fields, and unifying formats. Missing values were handled using forward-fill and backward-fill strategies or by replacing with appropriate defaults. Categorical variables were encoded numerically to ensure compatibility with machine learning models. The cleaning process was fully automated through Python scripts for reproducibility.

3 Problem Statement

The goal of this project is to understand which factors most significantly influence student performance and to build predictive models for student outcomes (e.g., exam scores and GPA). I focused on analyzing behavioral, academic, and contextual indicators to build models that can generalize to unseen student records and help support data-informed academic interventions.

4 Exploratory Data Analysis

Initial analysis revealed key relationships:

- **Positive correlations:** `Attendance`, `Hours_Studied`, and `Previous_Scores` showed the strongest associations with `Exam_Score`.
- **Statistical Test:** A t-test comparing students with attendance above and below 80% yielded a statistically significant difference in exam performance ($p < 0.001$).

I began my analysis by computing feature importances using a Random Forest Regressor implemented via `scikit-learn` to understand which features were most predictive of student performance. This revealed that `Attendance` and `Hours_Studied` were among the most important features. I supported this analysis by generating a correlation matrix using `pandas.corr()` and visu-

alized it with `seaborn.heatmap()` (Figure 3), as well as creating a feature importance bar graph (Figure 4).

To further examine key relationships, I used scatterplots generated with `seaborn.scatterplot()` and added best-fit regression lines using `numpy.polyfit()` and `matplotlib.pyplot.plot()` to visualize the relationship between `Attendance` and `Exam Score` (Figure 1), and `Hours Studied` and `Exam Score` (Figure 2). The respective slopes were 0.198 and 0.2893, indicating upward trends.

To statistically validate the impact of attendance on performance, I ran an independent two-sample t-test using `scipy.stats.ttest_ind()` to compare exam scores of students with less than 80% attendance to those with 80% or higher. The result was a p-value of less than 0.05, confirming that the difference in performance between low- and high-attendance groups was statistically significant.

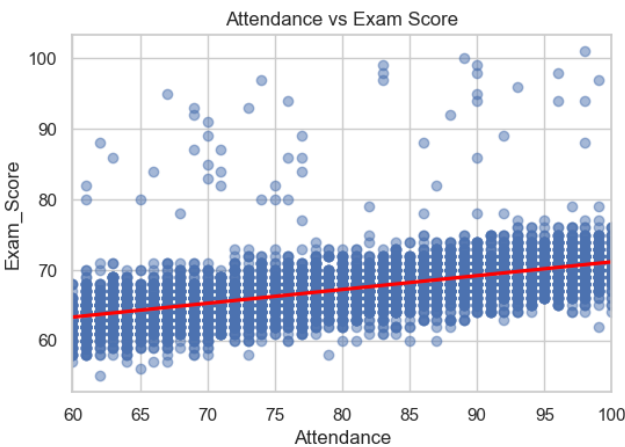


Figure 1: Attendance vs Exam Score

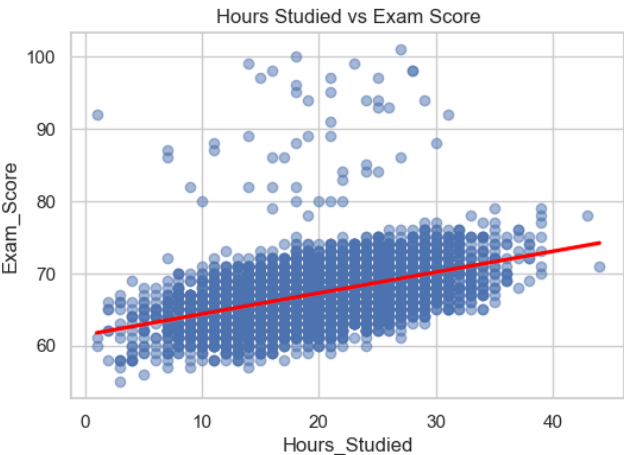


Figure 2: Hours Studied vs Exam Score

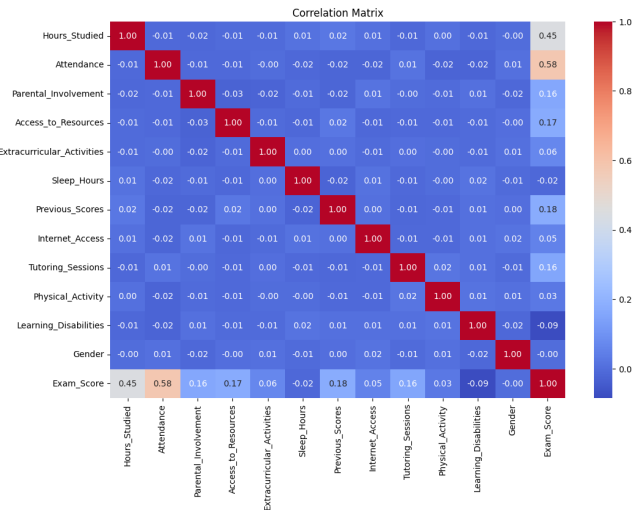


Figure 3: Correlation Matrix Heatmap

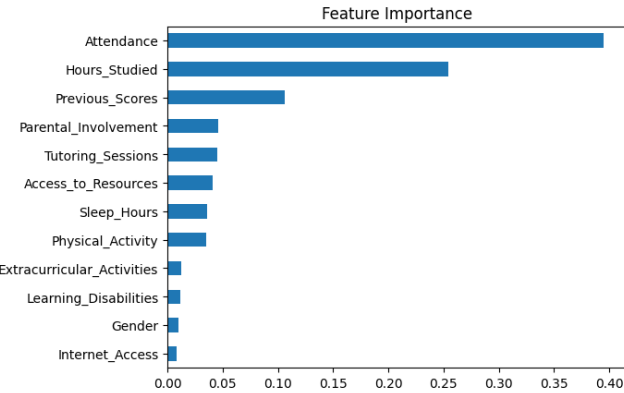


Figure 4: Feature Importances

5 Modeling and Prediction

Two supervised regression models were developed using Python’s `scikit-learn` library:

5.1 Linear Regression

- R^2 : 0.6680
- MAE: 1.16
- RMSE: 2.17

5.2 Gradient Boosting Regressor

- R^2 : 0.6448
- MAE: 1.27
- RMSE: 2.24
- Cross-validated R^2 : 0.5942 ± 0.0551

Linear Regression slightly outperformed the gradient model, indicating mostly linear relationships in the data.

5.3 Predicting GPA from Custom Input

To make the system practically usable, I built a regression model using three interpretable features: `StudyTimeWeekly`, `ParentalSupport`, and `Absences`. A Random Forest Regressor was trained using these features to predict GPA. I then created a minimal test file with the following input:

Student	StudyTimeWeekly	ParentalSupport	Absences
1	10.5	2	4
2	5.0	1	10
3	18.0	3	1

Table 1: Sample input used for GPA prediction using a Random Forest model

The model was able to predict GPAs for each student in the input file. The predictions were written to a new file, `predicted_gpa.csv`, which allows for downstream use in reporting, alerts, or integration with student dashboards.

To further assess model quality, I created a scatterplot comparing actual vs. predicted GPAs on the test set (Figure 5). This helped visualize how closely the predictions tracked with real data.

The final evaluation metrics for the Random Forest Regressor were as follows:

- R^2 : 0.8974

- MAE: 0.17
- MSE: 0.08
- RMSE: 0.28



Figure 5: Actual vs Predicted GPA

This demonstrates how the trained model can be deployed on new student data to provide real-time predictions based on minimal inputs, supporting early academic interventions.

6 Unsupervised Learning: Clustering

I applied KMeans ($k=3$) clustering using `scikit-learn`'s `KMeans` algorithm to identify behavioral profiles among students based on 8 features. Dimensionality reduction was applied using PCA to help visualize the clustering.

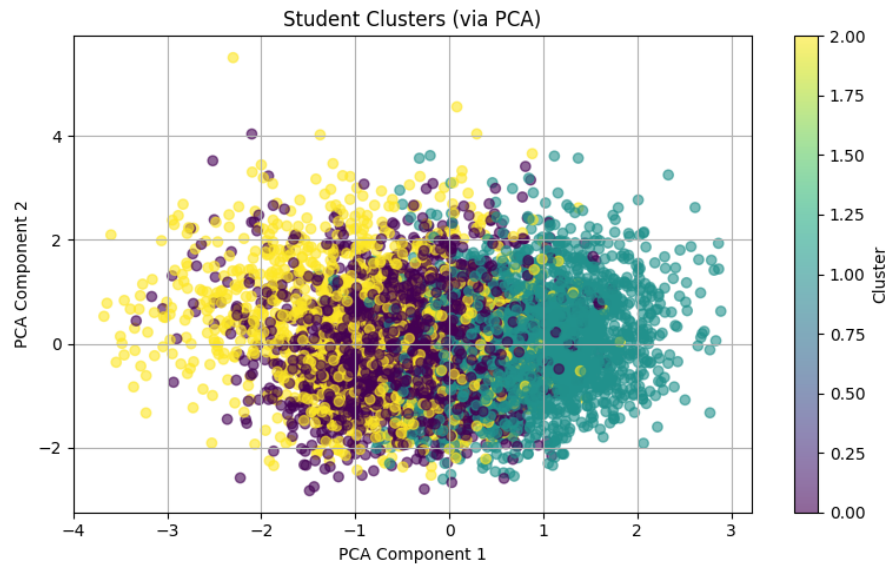


Figure 6: 2D PCA Visualization of Student Clusters

Cluster Interpretations

- **Cluster 0:** Moderate attendance, high study time, some learning challenges
- **Cluster 1:** Very high attendance, reliable study habits, low learning disabilities — “Model Students”
- **Cluster 2:** Lower attendance, highest proportion of students with learning disabilities — “Support Needed”

7 Conclusion

This project demonstrated that attendance, hours studied, and historical performance are key drivers of student success. From a modeling perspective, linear regression yielded strong performance, slightly outperforming gradient boosting in terms of RMSE and interpretability. Random forest regressors provided flexible, robust predictions even with minimal features and proved practical for real-time GPA estimation.

Clustering allowed for unsupervised behavioral segmentation, uncovering three distinct groups of students with unique academic profiles. These insights could support personalized educational interventions, such as targeting tutoring or parental engagement based on cluster membership.

Furthermore, the statistical tests confirmed that attendance has a significant impact on student outcomes, with performance differences between low- and high-attendance groups being statistically significant.

The work also illustrates how predictive models can be operationalized by accepting structured CSV inputs and outputting tailored predictions. With modest extensions such as model interpretability

(e.g., SHAP values) and richer longitudinal data, this framework could serve as the foundation for a full-scale academic early warning system.